



PERSONNEL IDENTITY VERIFICATION & SMART QR CODE SYSTEM

Comprehensive Client Operations Manual & Technical Reference

DOCUMENT METADATA

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1 Executive Summary & System Architecture

The **HGG Personnel Verification System** is an enterprise-grade digital authentication portal engineered to authenticate corporate credentials, eliminate physical badge counterfeiting, and provide verifiable proof of executive standing.

Each authorized employee ID card is printed with an official, high-density vector QR code linked to a dedicated, encrypted verification endpoint:

[https://hintergroupghana.com/verify/\[EMPLOYEE-ID\]](https://hintergroupghana.com/verify/[EMPLOYEE-ID])

When scanned with any standard smartphone camera or QR reader, the system immediately loads the official personnel record directly from the secure Sanity database in under 1 second without requiring any external app.

End-to-End Operational Workflow

Step 1: CMS Administration: Authorized managers create or update employee records in Sanity Studio.

Step 2: Secure QR Generation: The studio generates a tamper-resistant vector SVG QR code with the official HGG gold crest.

Step 3: Card Manufacturing: The vector QR code is printed directly onto commercial PVC identity badges.

Step 4: Instant Field Verification: Scanning the badge directs the scanner to the official live verification page, returning either a **Verified Active** identity or an **Unrecognized Record** security warning.

2 Sanity Studio Administration Guide

2.1 Accessing the Studio Portal

The administration portal is hosted at:

<https://hintergroupghana.com/studio>

Authorized administrators must sign in using approved corporate credentials via Google Workspace, GitHub, or direct Email/Password authentication.

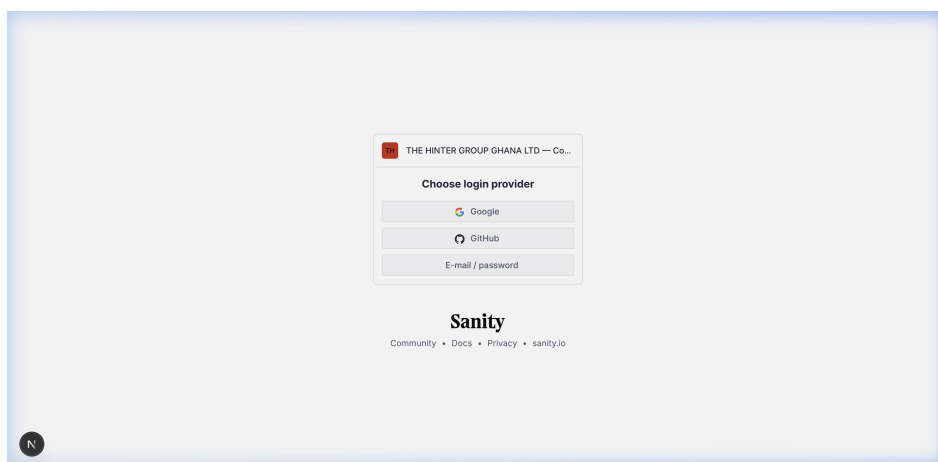


Figure 1: Sanity Studio Authentication Interface

2.2 Managing Employee Records

To create or modify an employee credential:

1. In the left-hand navigation pane, select "**Employee ID & Verification**".
2. Click the "**Create new**" icon (or select an existing record to edit).
3. Complete the required administrative fields:
 - **Employee ID**: Auto-generated with the next sequential ID in sequence (HGG-001, HGG-002). The system validates uniqueness and prevents duplicate assignments.
 - **Full Name**: Full legal name as printed on the official badge (e.g., *Charles N. Hammond*).
 - **Position / Official Role**: Corporate executive title (e.g., *Chairman & Chief Executive Officer*).
 - **Organization**: Defaults to THE HINTER GROUP GHANA LTD.
 - **Department / Division**: Corporate division (e.g., *Executive Leadership & Governance*).
 - **Verification Status**:
 - **Active / Verified**: Confirms valid identity with an emerald banner.
 - **Inactive / Expired**: Denotes lapsed credentials or former personnel.
 - **Suspended / Under Review**: Displays a red advisory indicating access revocation.
 - **Card Issuance Date**: Date of official issuance or badge printing.
 - **Approved Official Portrait**: High-resolution executive portrait image with interactive focal cropping.
 - **Internal Administrative Notes**: Private internal reference notes (never displayed on the public verification page).
4. Click "**Publish**" at the bottom right. Changes propagate globally across the verification network in real time.

3 QR Code Generation & Badge Manufacturing

Sanity Studio features an integrated **Employee QR Codes** management tool designed specifically for graphic designers and commercial badge printers.

3.1 Export Formats for Physical Badges

Inside the **Employee QR Codes** tab in the top studio bar, administrators can view live previews and export codes in two primary formats:

Export Format	Technical Specification	Recommended Application
Vector SVG	Scalable Vector Graphics with embedded HGG crest	Commercial PVC ID card printing, laser engraving, Adobe Illustrator / InDesign layouts. Lossless quality at any scale.
High-Res PNG	1200 × 1200 px raster with transparent background	Digital credentials, email badges, executive presentation summaries, PDF documents.

Table 1: Official QR Code Export Specifications

Printing Guidelines for Card Manufacturers

- **Minimum Physical Dimensions:** The QR code must be printed at no less than **20mm × 20mm** (0.8" × 0.8") to ensure 100% scan reliability under low-light conditions.
- **Contrast:** Maintain pure black-on-white or dark-navy on white background. Do not invert the QR modules.
- **Error Correction:** Built at **Level H** (30% error recovery), allowing the code to scan even with physical scratches or edge wear.

3.2 Official Example QR Code (Live Deployment Link)

Below is an authentic, production-ready example QR code generated with the HGG central shield emblem and metallic gold border frame. Scanning this QR code with any smartphone camera opens the live personnel credential record:

Direct Verification URL:

<https://hintergroup.netlify.app/verify/HGG-001>



Figure 2: Live Example QR Code · Scan to test with smartphone camera · Links to
<https://hintergroup.netlify.app/verify/HGG-001>

4 End-User Scanning Experience: Verified Identity

When a valid employee badge is scanned, the browser displays a luxury executive verification credential.

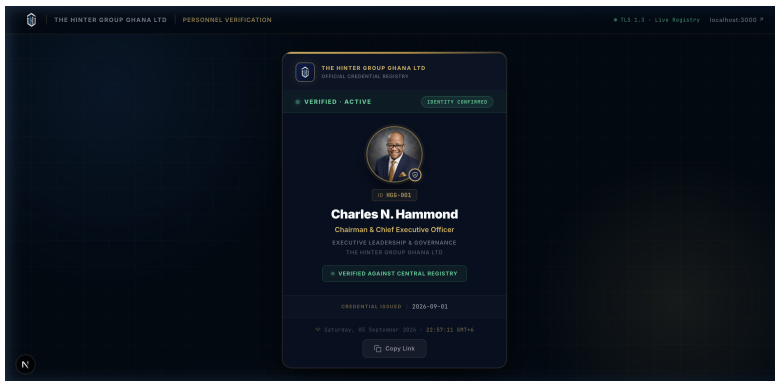


Figure 2: Verified Identity Page — Desktop View

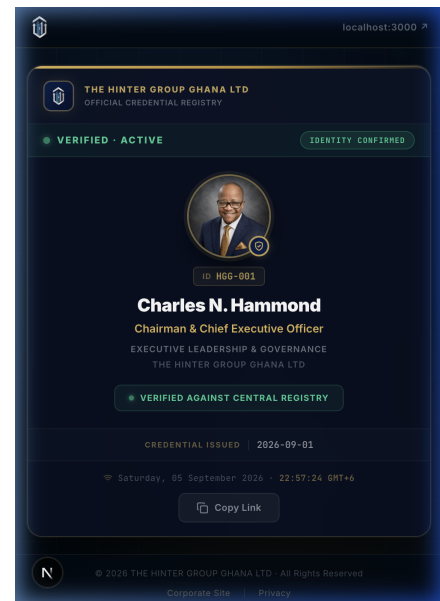


Figure 3: Mobile View

4.1 Component-by-Component Breakdown

- **Institutional Top Bar:** Displays the HGG logo, system designation ("*Personnel Verification*"), and a green pulsing TLS 1.3 · Live Registry indicator.
- **Organization Header:** Official HGG crest and "*Official Credential Registry*" designation.
- **Status Banner:** Pulsing emerald status dot with VERIFIED · ACTIVE and IDENTITY CONFIRMED security pill.
- **Biometric Portrait:** 104px circular portrait bordered in gold with an official gold verified shield emblem.
- **Employee ID Pill:** High-contrast monospaced badge displaying the unique identifier (e.g., HGG-001).
- **Executive Typography:** Bold full name, gold-accented corporate title, division, and organization name.
- **Central Registry Badge:** Direct confirmation pill affirming "*Verified against central registry*".
- **Credential Issue Date:** Synchronized issuance date matching the physical card.
- **Live Security Clock:** Real-time Greenwich Mean Time (GMT) clock updating every second. This confirms the verification is live and prevents fraudsters from presenting static screenshots.
- **Copy Verification Link:** One-click utility allowing border officials, corporate security, or banking partners to copy the authenticated verification URL.

5 End-User Scanning Experience: Unrecognized / Invalid Record

If an altered, revoked, or fabricated QR code is scanned, the verification portal immediately flags the credential with a **Credential Not Recognised** security alert.

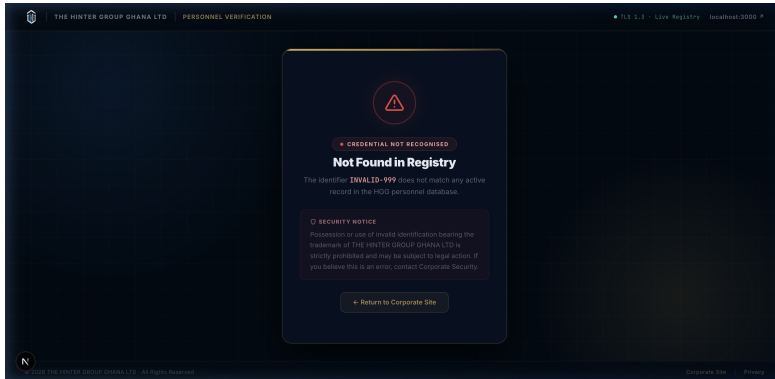


Figure 4: Unrecognized Record Warning — Desktop View

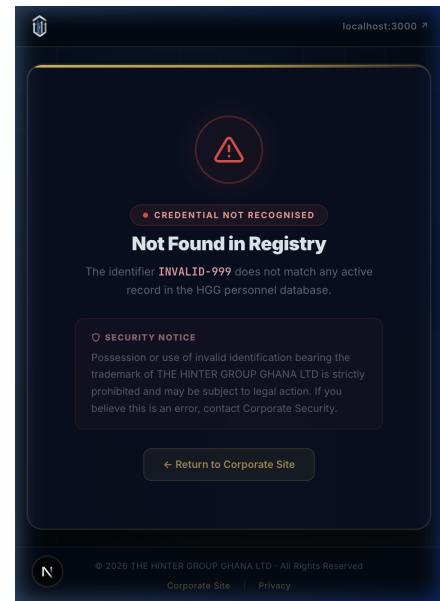


Figure 5: Mobile View

5.1 Security Response Components

- **Red Alert Shield:** Prominent red warning ring signifying authentication failure.
- **Status Pill:** Crimson alert badge reading CREDENTIAL NOT RECOGNISED.
- **Queried Code Display:** Quotes the scanned ID (*e.g.*, INVALID-999) and explicitly notes that no matching record exists in the HGG database.
- **Legal Advisory Notice:** Corporate warning stating that unauthorized possession or reproduction of credentials bearing the trademark of The Hinter Group Ghana Ltd is strictly prohibited and subject to legal prosecution.
- **Return Link:** Direct navigation button returning the user safely to the corporate site.

6 Security Architecture & Data Governance

Security Layer	Technical Implementation	Institutional Protection
Search Engine Privacy	robots.txt disallow rules + noindex, nofollow, noarchive, nosnippet meta tags	Prevents executive names, photos, and verification records from being indexed, cached, or scraped by public search engines.
Live Screenshot Prevention	Real-time JavaScript clock with GMT time synchronization	Guarantees that any presented verification page is actively connected to the server. Static screenshots are immediately identified by a frozen second hand.
Error-Resistant QR	ISO/IEC 18004 Level H (30% recovery) with vector HGG emblem	Badges remain 100% readable even if physically scratched, abraded, or partially obscured.
Zero-Cache Invalidation	Server-side dynamic fetching (revalidate: 0)	When an administrator changes a status in Sanity Studio, it takes effect world-wide within 1 second.
Transport Encryption	TLS 1.3 cryptographic handshake	End-to-end encryption protecting identity data between client devices and the registry servers.

Table 2: Cryptographic and Data Governance Architecture

Best Practices for Corporate Security Officers

- 1. **Annual Audits:** Regularly audit the Sanity Studio registry to ensure former staff or expired contracts are transitioned to `Inactive` status.
- 2. **Instant Revocation:** If an ID badge is lost or reported stolen, immediately switch the status to `Suspended` in Sanity Studio. Any subsequent scan of the physical badge will display the security alert screen.
- 3. **Badge Production:** Always provide the commercial printer with the official **Vector SVG** file downloaded directly from Sanity Studio.